



Zahraa Mohammed Alwan  
MSc. periodontics

## **Dental public health:**

It is "The science and art of preventing and controlling dental disease and promoting dental health through organized community efforts". It is that form of dental practice which serves the ***community*** as a one unit rather than an ***individual***.

**-The term dental public health is comparable and synonymous with community dentistry, community dental health, public health dentistry, social dentistry.**

## **Dental public health concerned with:**

- 1-Dental health education of the public.
- 2-Administration of programs of dental care.
- 3-Research and application of the research finding.
- 4-Prevention and control of dental diseases.

## **Public health impact of dental diseases:**

The most common diseases of the mouth are:

- 1-Dental caries.
2. Periodontal diseases.
3. Oral cancer
- 4.Developmental anomalies/acquired disorders of the oral structure.

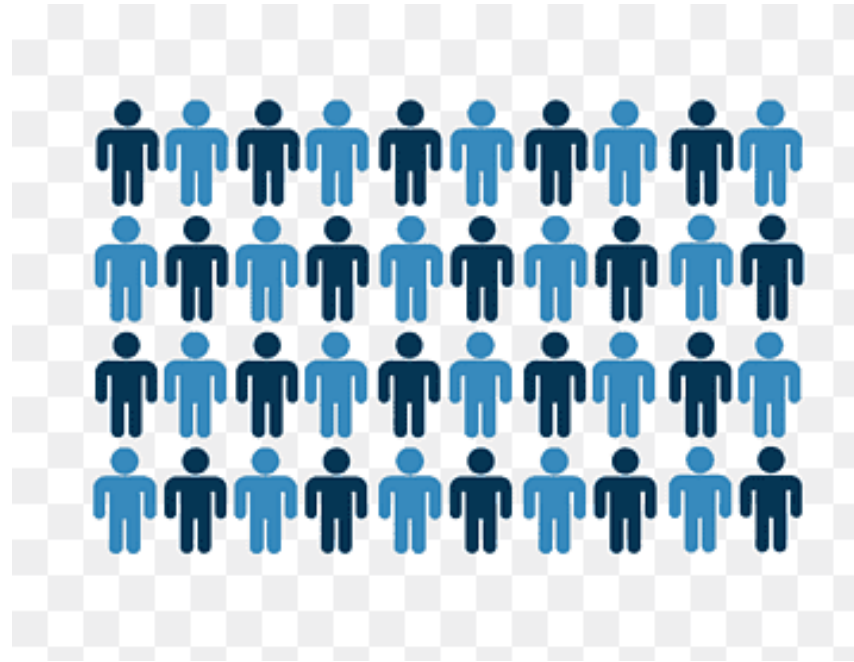


**Unique characteristics of these diseases are:**

- 1-They have universal prevalence.
- 2-They do not undergo remission or termination if left untreated.
- 3-They usually require more time and cost for treatment.

## Tools of dental public health:

**1-Epidemiology:** It is a study of distribution and determinants of health related events in population.



**2- Biostatistics:** Statistics is the science of compiling, classifying and tabulating numerical data and expressing the results in a mathematical or graphical form. Biostatistics is that branch of statistics concerned with mathematical facts and data relating to the biological events.



**3- Social Sciences:** Social sciences usually include sociology, cultural anthropology and psychology. The public health worker, when he embarks upon organized community effort, is very dependent upon the group behavior of the individuals, determined by their culture.

**4- Principles of Administration:** The dentist with a leadership role in public health program needs to know many of the principles by which large enterprises are administered, these are:

1- Organization: which deals with the structure of the agency and the arrangement of working people within it.

2- Management: which is concerned with the handling of personnel and operations in such a way that the work of the agency gets done.



**5- Preventive Dentistry:** It includes those practices by individuals and communities that improve oral health status. There are three levels of prevention. These are:

**1-Primary prevention:** This preventive measure is directed toward the pre-pathogenesis stage of the disease to prevent the occurrence of the disease (dental health education, oral hygiene measures, immunization, water fluoridation and fissure sealant).

**2-Secondary prevention:** It includes treatment of the disease to prevent the progression and recurrence of it (filling and pulp treatment).

**3-Tertiary prevention:** This preventive measure is directed toward the pathogenesis stage of the disease to restore the lost function, esthetics which has happened as a result of the disease (prosthesis).

## **Procedural steps in dental public health:**

1. **Survey:** Is the first step in the dental public health procedure. Surveys are methods for collection of data, in order to determine the amount of disease problems in a community.
2. **Analysis:** Information collected through a survey is subjected to an analysis in order to define specific health problems in the community. In the present modern world electronic data processing media as such as computers are resorted to for analyzing data. It is essential to obtain description of prevailing problem and their characteristics in order to achieve a correct diagnosis.

**3-Program planning:** After the problem and its characteristics are analyzed, the next step is program planning. During planning of the program; it should be taken in consideration: community's desire for intervention, cost and availability of proven interventions. The program must be accepted by the community, or to find an alternative method which is comprehensive. Hence it has to be ensured that the community is well informed about the programme.

**4- Ethics and planning approval:** All ethical clearance and approval from all the concerned persons and authorities is taken before implementing a public health program.

**5-Program operation:** After taking planning approval, a public health team has to be employed for executing the program.

**6-Financing:** Financing in public health programs are usually through the funds provided by the government or by the local authorities.

**7-Program appraisal:** This is the final step in any public health program where the effectiveness of the program is assessed. Data collected in initial surveys serves as a baseline against which effectiveness of an oral health program can be evaluated or assessed

## Differences between personal dental health & public health dentistry

| Characteristic            | Personal dental health                     | Dental public health                                             |
|---------------------------|--------------------------------------------|------------------------------------------------------------------|
| 1.Target                  | Individual patient                         | Community or group of individuals                                |
| 2.Visiting                | The patient comes to the practitioner      | The public health practitioner goes to the group of individuals  |
| 3. Major emphasis         | Restorative care                           | Preventive care                                                  |
| 4. Service provider       | Dentist alone, sometimes with an assistant | Health team professionals                                        |
| 5. Supportive disciplines | Psychology                                 | Sociology, psychology, education, epidemiology and biostatistics |
| 6. Perspective results    | Immediate                                  | Long term                                                        |
| 7. Funding                | By the patient                             | By government or local authorities                               |

